

AICA Arabic-First Multimodal RAG Strategy for Saudi Sovereign Enterprise Deployment

Executive Summary

AICA should treat **Arabic-first Multimodal RAG as a core architectural requirement, not an enhancement**. In Saudi enterprise and government environments, a large share of high-value “knowledge” is not clean, predominantly-English HTML. It is **Arabic PDFs, scanned circulars, forms, image-heavy reports, screenshots, and bilingual artifacts**. Under these conditions, **text-only RAG is structurally blind** to key evidence: page layout, tables, stamps/signatures, embedded figures, and the “visual semantics” that often carry the meaning. Research explicitly shows that text-only RAG cannot exploit visual cues in real-world multimodality documents, motivating **vision-language (VLM)-based RAG pipelines**. ¹

AICA’s defensible competitive position in Saudi Arabia is therefore: **sovereign Arabic-first grounded answering with audit-grade multimodal evidence**—including **page/region-level citations**, attached media, and permission-aware retrieval that is compatible with **PDPL** and national cybersecurity expectations (notably the **NCA ECC/CCC** control families). PDPL applies broadly to processing of personal data in the Kingdom and, in some scenarios, processing of personal data of individuals residing in the Kingdom by parties outside the Kingdom, and it defines “processing” very broadly (including storing, retrieving, disclosing, and transmitting). ² The NCA’s ECC and CCC set baseline cybersecurity requirements for national entities (and encourage broader adoption), including expectations for continuous compliance, logging/monitoring, incident management, cryptography, and cloud/hosting controls. ³

Strategic position (opinionated):

- Multimodal RAG is **critical** for AICA’s credibility in Saudi regulated and semi-regulated sectors because the most defensible answers require **evidence objects** (page snippets, screenshots, tables, forms, audio timestamp clips), not just paraphrased text; multimodal retrieval materially reduces “Arabic OCR brittleness” and “layout loss.” ⁴
- AICA should adopt a **dual-layer retrieval architecture**: (1) best-in-class Arabic text retrieval (hybrid lexical+dense, Arabic-aware normalization, Arabic reranking), and (2) **vision-first page retrieval for visually rich and scanned documents** (e.g., ColPali-like page embeddings) with explicit linkage to OCR/layout outputs. This is more robust than “OCR-only + text embeddings,” which fails on low-quality scans and layout-dependent meaning. ⁵
- For Saudi sovereign hosting, AICA should prioritize **self-hostable** components for search, orchestration, evaluation, and monitoring, and treat vendor components as **swappable modules** to avoid lock-in. (Examples of self-hostable search foundations include Elastic self-managed deployments; open-source vector+hybrid approaches exist in OpenSearch.) ⁶

Key recommendations (practical):

- Build an **Arabic-first Multimodal Document Representation** (“document object model”) that preserves page layout, bounding boxes, reading order, tables, figures, and links every extracted claim back to a stable evidence anchor (doc_id, page, bbox / timestamp). The goal is “audit-grade grounded answers,” not just “chat.” ⁷
- Use **hybrid retrieval** (BM25 + dense vectors) for Arabic text plus **VLM page retrieval** for scanned/ layout-heavy PDFs, then fuse results with a deterministic rank-fusion layer and an Arabic reranker. ⁸
- Treat Arabic as a first-class subsystem: Arabic OCR quality controls, Arabic normalization and morphology-aware query expansion, Arabic embeddings and reranking, Arabic citation UX, and bilingual Arabic↔English bridging that works even when documents are mixed-language. (This is where “supports Arabic” typically collapses into “Arabic works poorly.”) ⁹
- Build governance to match Saudi expectations: PDPL-aware data minimization and storage controls; lifecycle transparency, privacy/security-by-design and decision traceability aligned with Saudi Data and Artificial Intelligence Authority (SDAIA) ¹⁰ AI Ethics Principles; cybersecurity logging, incident response, encryption/key management aligned to National Cybersecurity Authority (NCA) ¹¹ ECC/CCC. ¹²

Research Scope and Method

This research focused on **Arabic-first multimodal retrieval + enterprise assistant delivery under Saudi sovereignty constraints** rather than generic RAG templates.

Research approach

- Multimodal RAG and document AI: prioritizing recent peer-reviewed research and widely cited preprints on **VLM-based retrieval**, visually rich document benchmarks, and OCR-free / layout-aware document understanding (e.g., ColPali, VisRAG, LayoutLMv3, Donut, DocVQA, DocLayNet). ¹³
- Arabic-first components: Arabic OCR limitations and failure modes; Arabic embeddings and Arabic language model foundations (e.g., AraBERT; multilingual embeddings). ⁹
- Saudi policy and controls: PDPL guidance and national cybersecurity control frameworks (ECC/CCC) and Saudi Cloud First policy statements relevant to data localization expectations. ¹⁴
- Competitive intelligence: competitor documentation for language support, grounding, connectors/permissions, and deployment models, with special attention to **Arabic limitations** and **sovereign hosting feasibility**. ¹⁵

Source categories (priority order) 1) Official documentation and standards-like government publications (PDPL guidance, NCA ECC/CCC, Cloud First policy). ¹⁴

2) Research papers and benchmarks (ArXiv, ACL/NAACL, OpenReview, CVF). ¹⁶

3) Vendor technical docs (language support, grounding behavior, connectors/permissions, deployment models). ¹⁷

Limitations and uncertainty

- “Arabic support quality” is rarely measured publicly as **Arabic-first enterprise usability**, so competitor comparisons often require separating **documented feature support** (high confidence) from **performance/quality claims** (lower confidence) unless benchmarked. ¹⁸
- Some Saudi government guideline PDFs were not directly retrievable from the primary domain due to access controls; where possible, this report relies on accessible official alternatives (e.g., PDPL guidance PDF and AI ethics principles PDF). ¹⁹
- Multimodal RAG remains fast-moving; the architecture recommendations emphasize **modularity** so AICA can swap retrievers, OCR engines, and VLMs without redesigning the whole product. ²⁰

Observed Facts About Multimodal RAG

This section is **facts-only** (no recommendations), grounded in cited sources.

Multimodal RAG exists because text-only RAG is insufficient for real documents

- VisRAG explicitly motivates multimodal RAG by stating that current RAG systems are “solely based on text,” making it **impossible to utilize vision information like layout and images** that are crucial in real-world multi-modality documents; VisRAG proposes a VLM-based RAG pipeline to address this. ²¹
- ColPali introduces a VLM trained to produce **multi-vector embeddings from images of document pages**, paired with late-interaction matching; it reports outperforming “modern document retrieval pipelines” while being “drastically simpler, faster and end-to-end trainable.” ²²
- Donut (OCR-free Document Understanding Transformer) argues OCR-dependent pipelines suffer from **OCR cost, inflexibility across languages/document types**, and **OCR error propagation**; it proposes an OCR-free VDU approach and reports strong performance on document understanding tasks. ²³

Layout-aware document understanding is an established research direction

- LayoutLMv3 is a multimodal Transformer for Document AI with unified text+image masking and word-patch alignment to learn cross-modal alignment, targeting tasks including form understanding, receipt understanding, and document visual question answering. ²⁴
- DocLayNet is a large human-annotated layout segmentation dataset (80,863 pages) designed for diverse corporate document layouts and 11 layout classes, enabling evaluation/training for layout analysis. ²⁵
- DocVQA provides a benchmark dataset for question answering on document images (tens of thousands of QA pairs), supporting evaluation of document-image understanding. ²⁶

Arabic OCR is intrinsically challenging and quality varies

- A review of Arabic OCR highlights Arabic script properties that complicate recognition (including cursive writing, dots/diacritics, right-to-left direction, and morphological richness). ²⁷

- A comparative OCR study (including Tesseract, Amazon Textract, and Google Document AI) explicitly notes that the Arabic corpus was not submitted to Textract because Textract does not support Arabic; it evaluated Arabic OCR using other engines. ²⁸
- Public vendor documentation shows uneven Arabic OCR support: Google Cloud Vision OCR lists Arabic (ar) as supported. ²⁹ Microsoft Azure OCR documentation states Arabic is among supported languages for printed text. ³⁰ ABBYY documentation states Arabic is supported and that “end-to-end recognition for Arabic” is available to improve quality on low-quality images. ³¹

Arabic NLP foundations exist, but Arabic-first quality is not automatic

- AraBERT is a transformer-based model pre-trained specifically for Arabic; the paper characterizes Arabic as morphologically rich and reports strong performance across Arabic NLP tasks compared to baselines. ³²
- Multilingual E5 text embedding models are open-source and evaluated in a technical report; documentation also cautions that low-resource languages may see performance degradation. ³³

Arabic speech recognition exists, but must handle dialects and code-switching

- Whisper is trained on 680,000 hours of multilingual/multitask supervised data and supports multilingual speech recognition; external evaluation work notes Whisper demonstrates strong capabilities across multiple languages including Arabic (though Arabic labeled hours are limited relative to total training). ³⁴

Saudi governance and cybersecurity controls are concrete and operationally relevant

- PDPL guidance states PDPL applies to processing of personal data in the Kingdom and, in certain cases, processing of personal data of individuals residing in the Kingdom by parties outside the Kingdom; it defines personal data broadly and defines “processing” to include collecting, storing, retrieving, using, disclosing, transmitting, publishing, erasing, etc. ²
- The NCA ECC 2-2024 document states it provides minimum cybersecurity requirements for national entities within scope, emphasizes continuous compliance, and includes domains and controls covering cryptography, event logs/monitoring, incident management, backup/recovery, etc. ³⁵
- The NCA CCC 2-2024 document (as an extension to ECC) defines minimum cybersecurity requirements for cloud computing for CSPs and CSTs and is explicitly framed in terms of protecting confidentiality/integrity/availability. ³⁶
- Saudi Cloud First policy documentation states that all data in Government Cloud and Commercial Governmental Cloud should be geographically located inside the borders of Saudi Arabia, and emphasizes cybersecurity evaluation governed by NCA-issued regulations. ³⁷

Competitor language support and Arabic readiness varies materially (documented)

- Microsoft ³⁸ documentation states Microsoft 365 Copilot supports Arabic prompts/responses (language support list includes Arabic). ³⁹
- Amazon Web Services (AWS) ⁴⁰ documentation states Amazon Q Business is optimized to respond in English and only indexes English documents; it recommends indexing only English content. ⁴¹
- Glean ⁴² documentation states search results are a subset of documents written in the same language as the query, and cross-language answering (e.g., English query needing Spanish doc) is “not GA.” ⁴³

- Google Cloud ⁴⁴ documentation for Vertex AI Search lists Arabic (ar-001) among supported languages and indicates feature support varies; grounded generation APIs are listed as supported for Arabic. ⁴⁵

Why Multimodal RAG Matters for AICA

Strategic relevance in Saudi enterprise and government reality

Saudi organizations (particularly in regulated and semi-regulated environments) typically require **auditability, provenance, and defensible evidence**. That requirement is incompatible with a retrieval strategy that treats knowledge as “just text chunks.” When an answer is derived from a scanned circular, a stamped letter, a table inside a PDF, or a screenshot in a report, the **evidence object** (page region, extracted table, diagram) is often the only trustworthy artifact. VisRAG explicitly frames the problem: text-only RAG cannot use layout/images that matter in real documents. ²¹

Why text-only RAG is insufficient for AICA (especially Arabic-first)

Text-only RAG breaks in specifically Saudi/Arabic contexts due to a compounding set of failure modes:

- 1) **OCR brittleness in Arabic**, especially on low-quality scans, mixed fonts, stamps, rotated pages, and form-like templates; Arabic OCR is inherently challenging because of script characteristics and diacritics that can change meaning. ⁴⁶
- 2) **Layout-dependent semantics**: many “answers” live in a table cell, a header/footer, a signature block, a stamp, or a diagram. The meaning is partly visual, not purely lexical. ColPali and VisRAG exist because modern systems still struggle to use visual cues effectively for document retrieval, motivating VLM-based retrieval for visually rich documents. ⁴⁷
- 3) **Arabic morphology and business phrasing**: Arabic is morphologically rich; enterprises also use code-switching and loanwords. Arabic-specific models like AraBERT exist because Arabic language understanding differs materially from English. ³²
- 4) **Bilingual corpora**: Saudi enterprises commonly store Arabic-first documents with English technical annexes, or bilingual policies; naïve “same-language retrieval only” becomes an operational failure (not a feature gap). Glean’s own documentation effectively acknowledges this limitation by stating cross-language answering is not GA. ⁴³

Inferences for AICA (explicit, not presented as observed facts)

The following are **inferences** derived from the observed facts and the Saudi product constraints:

- If AICA is “Arabic-first,” then **Arabic OCR, Arabic retrieval, and Arabic evidence UX** must all be treated as *primary KPIs*; otherwise, the assistant will appear to “support Arabic” while failing in the most important Saudi deployments (policies, circulars, and internal directives). ⁴⁸

- Because Arabic OCR will remain imperfect, AICA needs a **second retrieval signal that bypasses OCR** for visually rich documents—e.g., page-image embeddings as in ColPali-like retrieval. This is not optional in a Saudi enterprise assistant that must answer from scanned PDFs at scale. ⁴⁹
- Sovereign hosting expectations plus Cloud First localization expectations imply AICA's differentiation must include **deployability inside Saudi-controlled infrastructure** with auditable security controls aligned to ECC/CCC and PDPL constraints; SaaS-only competitors will remain strategically constrained even if feature-rich. ⁵⁰

Best Practices for Enterprise AI Conversational Assistants

This section focuses on enterprise-grade practices relevant to Saudi environments and Arabic-first delivery.

Grounding, trustworthiness, and evidence presentation

Enterprise assistants must minimize hallucination and maximize verifiability. Google's grounding documentation describes grounding as connecting model output to verifiable data sources to reduce hallucinations, particularly where accuracy and reliability matter. ⁵¹ In ServiceNow AI Search, Now Assist explicitly frames grounding by constraining the context passed to the LLM to increase likelihood that responses are grounded in indexed content. ⁵²

For AICA, "citations" must go beyond "document title" and support **page and region anchoring** (page number + bounding box) for PDFs and **timestamp anchoring** for audio/video. This aligns both with enterprise audit needs and with Saudi AI ethics expectations that decisions be traceable and explainable. ⁵³

Permission-aware retrieval and enterprise security controls

Competitors emphasize the importance of permission enforcement at ingestion and query time:

- Microsoft 365 Copilot architecture documentation states Copilot uses Microsoft Graph to access user data in the user's unique context, including documents the user has permission to access. ⁵⁴
- Glean connector documentation emphasizes connectors fetch both content and permissions maps to ensure results adhere to source permissions. ⁵⁵

For Saudi deployments, permissioning is not just a product feature—it is a compliance and trust requirement. Architecturally, permission checks must be applied **before** generation and must also control **which evidence objects** can be attached or previewed.

Governance, auditability, and alignment to Saudi expectations

Saudi AI ethics guidance (SDAIA AI Ethics Principles) explicitly emphasizes privacy/security-by-design, transparency/traceability, and lifecycle governance (plan & design → data preparation → build & validate → deploy & monitor). ⁵⁶ NCA ECC and CCC provide baseline cybersecurity expectations, including (as examples) event logs/monitoring management, incident and threat management, cryptography and key management, and cloud/hosting cybersecurity controls. ⁵⁷ PDPL guidance makes clear that storing, retrieving, disclosing, and transmitting personal data are all "processing," which means the assistant's logs,

traces, and training/evaluation datasets can become regulated personal data stores if not designed carefully. ²

Best-practice implication for AICA: governance must cover ingestion pipelines, model prompts, retrieval traces, user feedback loops, and escalation workflows, with data classification and retention policies applied end-to-end.

Evaluation and observability as production requirements (not optional)

Enterprise-grade assistants require systematic evaluation loops. Ragas is explicitly designed to move from ad-hoc “vibe checks” to systematic evaluation of RAG applications, and the RAGAS paper proposes reference-free evaluation dimensions. ⁵⁸ TruLens introduces the “RAG triad” style evaluations (context relevance, groundedness, answer relevance). ⁵⁹ Arize Phoenix is an open-source observability platform for tracing and evaluation using OpenTelemetry-based instrumentation. ⁶⁰

For AICA specifically, evaluation must be **Arabic-first and multimodal-aware**:

- OCR evaluation must be part of the pipeline because Arabic OCR failures propagate directly into retrieval failures. ⁶¹
- Multimodal “evidence correctness” must be evaluated: not just whether the answer is correct, but whether the attached page region/time clip actually contains the claimed evidence.

Arabic-first usability and failure patterns to avoid

AICA must avoid the common trap: “Arabic UI + multilingual embeddings + generic OCR = Arabic product.” Arabic-first requires:

- consistency of terminology and named entities across Modern Standard Arabic and business phrasing (often with code-switching),
- robust handling of Arabic orthographic variation and morphological forms,
- evidence UX that feels trustworthy for Arabic readers (clear citations, consistent formatting, predictable quoting). ⁶²

A warning sign in competitor ecosystems is explicit “same-language only” search/answer behavior (e.g., Glean’s language support statement), which is directly misaligned with bilingual Saudi corpora. ⁴³

Competitive Market Landscape

Market Overview

The market is converging on “enterprise copilots,” but the underlying capabilities split into distinct layers: 1) **Enterprise assistant UIs** embedded in productivity suites or platforms (Copilot, ServiceNow). ⁶³

2) **Enterprise search + RAG platforms** with connectors and permission models (Glean, Coveo, Sinequa). ⁶⁴

3) **Search infrastructure** that organizations self-host for lexical + vector + multimodal retrieval (Elastic, OpenSearch). ⁶

4) **Document AI/OCR and multimodal retrieval research** increasingly shifting retrieval from “OCR text” to “page image embeddings” for visually rich documents (ColPali, VisRAG; Donut as OCR-free). ⁶⁵

For AICA, the Saudi-specific wedge is: **Arabic-first + sovereign deployment + multimodal evidence UX**.

Competitor Profiles

Microsoft ³⁸ enterprise copilots

Microsoft 365 Copilot is documented as an AI tool whose responses can include work content that users have permission to access; its architecture relies on Microsoft Graph to access the user’s context and permitted data. ⁶⁶ Microsoft documentation also lists Arabic among supported prompt/response languages for Microsoft 365 Copilot. ³⁹

Multimodal implications (documented vs inferred): Microsoft’s ecosystem can access documents and generate responses; public docs emphasize grounding and permissions via Graph, but do not, in the cited sources, specify audit-grade page-region evidence UX for scanned Arabic PDFs. (That gap is a potential AICA differentiation; it is an inference, not a documented claim.) ⁶⁷

Saudi suitability: Strong enterprise footprint; however, sovereign “LLMs hosted locally inside Saudi by the customer” is not the default posture of suite copilots (inference based on deployment model norms, not a cited product limitation).

Google Cloud ⁴⁴ Vertex AI Search and grounding

Google’s documentation describes grounding as connecting model output to verifiable sources to reduce hallucinations, and supports grounding with Vertex AI Search. ⁵¹ Vertex AI Search language documentation lists Arabic (`ar-001`) as supported, with grounded generation APIs supported (feature support varies). ⁶⁸

Multimodal relevance: Vertex AI Search documentation includes “media search” and grounded generation APIs, with language support varying per feature. ⁴⁵ For Saudi sovereign constraints, the main friction is that typical usage is managed cloud APIs; a fully sovereign deployment requires isolation and local hosting choices (AICA’s core requirement).

Amazon Web Services (AWS) ⁴⁰ Amazon Q Business

AWS documentation is explicit: Amazon Q Business is optimized for English and **only indexes English documents**. ⁴¹ AWS documentation also lists extensive connector support (data sources), implying mature enterprise integration—yet the English-only indexing constraint is a direct mismatch to Arabic-first knowledge bases. ⁶⁹

Bottom line: strong enterprise integration story, but **Arabic-first retrieval is structurally blocked** unless customers implement non-native workarounds (translation pipelines, custom ingestion) outside Q Business itself (inference from English-only indexing statement). ⁷⁰

Glean ⁴² **enterprise search + assistant**

Glean documentation emphasizes connectors that parse content and permissions, enforcing access controls. ⁷¹ Glean language support documentation states search results are restricted to documents in the same language as the query; cross-language answering is not GA. ⁴³ Glean also documents two deployment models: SaaS hosted and “Customer-Hosted (Cloud-Prem)” where Glean deploys its tenant as a managed service in the customer’s cloud environment; it explicitly states this is not a traditional self-host model and Glean retains operational control access. ⁷²

Saudi fit: this deployment flexibility can help data residency, but AICA’s strict “all models and optimization hosted locally” requirement goes further than “managed in customer VPC,” and Glean’s same-language limitation is misaligned with bilingual Saudi corpora unless mitigated via upstream translation or bilingual embeddings (inference). ⁷³

ServiceNow ⁷⁴ **Now Assist / AI Search RAG**

ServiceNow documentation states Now Assist in AI Search combines search with an LLM and that constraining context passed to the LLM increases likelihood responses are grounded in indexed content. ⁷⁵ ServiceNow also documents its RAG application conceptually (RAG enhances search accuracy by limiting LLM focus to contextual dataset). ⁷⁶

Saudi fit: strong workflow embedding in ITSM and enterprise processes; actual Arabic-first quality and sovereign deployment specifics are not established in the cited sources (uncertain).

Coveo ⁷⁷ **generative answering on enterprise index**

Coveo documentation states its Relevance Generative Answering model generates answers “based solely on the content that you specify,” residing in a secure index. ⁷⁸ Coveo positioning also emphasizes citations and multilingual answers (marketing-oriented), but the core documented value is constraining generation to indexed content. ⁷⁹

Saudi fit: strong relevance/search heritage; sovereign “Saudi-local hosting including LLMs” is not established in cited docs (uncertain).

Sinequa ⁸⁰ **(ChapsVision) sovereign/regulated enterprise search posture**

Sinequa’s own materials (ChapsVision site) explicitly claim deployability **on-prem, private cloud, or air-gapped**, and emphasize governance and traceability. ⁸¹ This is closer to Saudi sovereign constraints than SaaS-only competitors, at least at the deployment model level (documented claim).

Arabic-first: multilingual claims exist, but Arabic-first quality is not demonstrated in the cited sources (uncertain).

Elastic ⁸² self-managed search and multimodal retrieval building blocks

Elastic documentation supports self-managed on-prem deployments. ⁸³ Elastic also publishes technical guidance on multimodal image retrieval using embeddings (e.g., CLIP/SigLIP) and kNN vector search in Elasticsearch. ⁸⁴

Saudi fit: strong because it is self-hostable and acts as infrastructure; however, AICA would need to build the Arabic-first multimodal ingestion, permissioning, and assistant orchestration layers on top.

IBM ⁸⁵ watsonx / conversational search posture and Saudi Arabic LLM ecosystem

IBM content positions watsonx Assistant as leveraging retrieval from documents for conversational answers (RAG), and IBM provides tutorials for building RAG systems. ⁸⁶ Importantly for the Saudi market, IBM's newsroom states that SDAIA's Arabic LLM "ALLaM" launched on watsonx, described as open-source (with optional AUP). ⁸⁷

This signals that Arabic-first foundation models are becoming enterprise-consumable in the region, and it raises the bar for AICA to be **at least as strong as top Arabic LLMs** when deployed locally.

Competitor Comparison Matrix

The matrix below separates **documented facts** (explicit in cited sources) from **inferred gaps** (marked as inference). "Arabic-first quality" is not the same as "Arabic supported"; where quality is not documented, it is listed as **Unknown**.

Competitor / Option	Arabic prompts/ responses (documented)	Arabic indexing / retrieval (documented)	Cross-language retrieval (Arabic↔English)	Multimodal ingestion breadth (docs/ images/audio/ video)	Scanned PDF robustness	Evidence UX (page/region/ timestamp)
Microsoft 365 Copilot	Yes ⁸⁸	Not specifically documented in cited sources	Unknown	Documents via Graph context (documented), true multimodal specifics not shown	Unknown	Unknown
Google Vertex AI Search + grounding	Arabic supported for search/ grounded gen APIs ⁶⁸	Yes (Arabic supported) ⁶⁸	Unknown	Media search exists; language support varies by feature ⁶⁸	Unknown	Grounding concept documented; evidence UX specifics vary (unknown) ⁹⁰

Competitor / Option	Arabic prompts/ responses (documented)	Arabic indexing / retrieval (documented)	Cross-language retrieval (Arabic↔English)	Multimodal ingestion breadth (docs/ images/audio/ video)	Scanned PDF robustness	Evidence UX (page/region/ timestamp)
Amazon Q Business	English-optimized ⁷⁰	English-only indexing ⁷⁰	Not applicable / constrained	Connectors strong (documented) ⁹²	Weak for Arabic corpora	Citations exist in product positioning, but Arabic corpus blocked
Glean	Chat matches query language ⁴³	Same-language subset behavior ⁴³	Not GA ⁴³	Many enterprise sources & file types (documented) ⁷¹	Unknown	Unknown
ServiceNow Now Assist in AI Search	Not documented	Not documented	Unknown	Enterprise platform search + LLM (documented) ⁷⁵	Unknown	Grounding via constrained context (documented) ⁷⁵
Coveo RGA	"Multilingual" marketing (not a guarantee)	Depends on indexed content	Unknown	Strong search heritage; multimodal specifics unknown	Unknown	Answer based solely on specified content (documented) ⁷⁸
Sinequa	Not documented	Multilingual claims (not Arabic-first proven)	Unknown	Enterprise search & "agentic AI" positioning	Potentially strong (inference from deployment + enrichment posture)	Governance/ traceability emphasized (documented) ⁹⁵

Competitor / Option	Arabic prompts/ responses (documented)	Arabic indexing / retrieval (documented)	Cross-language retrieval (Arabic↔English)	Multimodal ingestion breadth (docs/ images/audio/ video)	Scanned PDF robustness	Evidence UX (page/region/ timestamp)
Elastic-based custom build	N/A	Depends on your pipeline	Yes if you build it	Image search via embeddings documented; full multimodal pipeline is DIY ⁹⁶	Potentially strong if paired with VLM page retrieval	Can be audit-grade if built
OpenSearch-based custom build	N/A	Depends on your pipeline	Yes if you build it	Supports BM25 + vector + “multimodal image search” framing ⁹⁸	Potentially strong if paired with VLM page retrieval	Can be audit-grade if built
AICA (recommended target)	Arabic-first by design	Arabic-first hybrid + VLM page retrieval	Yes (Arabic↔English bridging)	Text + PDFs + embedded images + standalone images; audio/video via ASR	High (OCR + VLM retrieval)	Page/region/ timestamp citations + attached evidence

Pros and Cons by Competitor (condensed, decision-useful)

- Microsoft 365 Copilot: strong enterprise integration and documented Arabic prompt support; but sovereignty + Arabic-first evidence UX for scanned PDFs is not established in available sources, and the product is suite-centric rather than Saudi-sovereign-centric. ⁶⁷
- Google Vertex AI Search: strong grounding story and Arabic language support; typical deployment is managed cloud which conflicts with strict local hosting; also multimodal and Arabic feature parity varies. ¹⁰⁰
- Amazon Q Business: excellent connector ecosystem but English-only indexing is a decisive blocker for Arabic-first knowledge assistants. ¹⁰¹
- Glean: well-articulated permissions model and deployment options (SaaS and cloud-prem), but same-language constraint is strategically misaligned with Saudi bilingual corpora, and it is not full sovereign self-host. ⁷³
- ServiceNow: strong workflow embedding and explicit RAG grounding concept in AI Search; Arabic-first capabilities and sovereign posture are unclear from cited sources. ⁵²
- Coveo: strong “generate only from specified content” discipline; less clarity on sovereign deployment and Arabic-first depth. ¹⁰²
- Sinequa: explicitly positions for sovereignty and air-gapped deployment; Arabic-first depth still must be proven via evaluation. ⁸¹

- Elastic/OpenSearch infrastructure builds: strongest for sovereignty and customization; require AICA to build major product layers (connectors, Arabic-first multimodal ingestion, orchestration, governance UX). ¹⁰³

Competitive Gaps and White Space

The most defensible white space for AICA is the intersection of:

- **Arabic-first** (not merely Arabic-supported),
- **multimodal evidence grounding** (page-region citations, table extraction, diagram capture),
- **sovereign Saudi-local deployment** (models + retrieval + evaluation inside Saudi-controlled infrastructure),
- **government/regulated procurement readiness** (auditability, security controls alignment). ¹⁰⁴

This combination is weakly served by cloud-native copilots and by generic “multilingual RAG” platforms.

What AICA Must Do to Compete

AICA must be able to win head-to-head in the **hardest Saudi knowledge cases**:

- “Show me the clause in the scanned circular that authorizes X” (requires page-region evidence).
- “Explain this table in the PDF and cite the cell values” (requires table structure).
- “What did the speaker say at minute 12:40 in the training video about procedure Y?” (requires ASR + timestamp citation).
- “Find the screenshot that shows KPI Z threshold and attach it.” (requires image retrieval + evidence packaging). ¹⁰⁵

Recommended Multimodal RAG Solution for AICA

Product Vision

AICA should be positioned as: **the Saudi sovereign, Arabic-first, evidence-grounded enterprise assistant** that can answer questions from messy real-world knowledge assets (scanned PDFs, bilingual policies, image-heavy reports, audio/video) while producing **audit-grade citations and attachments**.

AICA’s product promise should be measurable:

- “Every factual claim must be backed by at least one evidence anchor.”
- “If evidence is insufficient or permission-protected, AICA refuses or asks for clarification.”
- “Arabic-first: Arabic query understanding, Arabic retrieval quality, Arabic OCR quality, Arabic answer generation, Arabic citation UX.”

This vision directly aligns with AI ethics expectations around transparency and traceability and with PDPL’s broad definition of regulated “processing,” which implies careful control of what is stored, logged, and retained. ¹⁰⁶

Target User Experience

User flow (Arabic-first): 1) User asks in Arabic. 2) AICA detects intent + modality need: “text-only answer” vs “answer with evidence attachments.” 3) AICA retrieves evidence across modalities, filtered by permissions. 4) AICA responds primarily in Arabic with:

- a structured answer,
- numbered citations (doc title + page + highlighted snippet),
- optional attached evidence objects (page screenshot crop, table extract, image, audio clip with timestamp),
- a “confidence & evidence sufficiency” indicator explaining *why* the evidence is sufficient or insufficient.

Evidence UX requirement: Every attachment is introduced with a short Arabic caption (“مقتطف من الصفحة ٣...”) and the UI supports tapping to open the original PDF at the cited page with highlights.

This maps to the “decision traceability” and “transparent, explainable rationale” emphasis in AI ethics principles. ⁵⁶

End-to-End Architecture

Opinionated architectural stance: AICA should implement **modality-specific indexes with a unified evidence model**, not a single “unified embedding store” that pretends all modalities are equivalent.

Reason: ColPali-style retrieval demonstrates that page-image embeddings with late interaction can outperform typical text pipelines for visually rich documents—but they are operationally different from text chunk retrieval. ²²

Core layers

- Ingestion & normalization layer: produces a canonical document object model.
- Multi-index retrieval layer:
 - Text index: lexical + dense vectors.
 - Page-image index: VLM page embeddings (for scanned/visually rich PDFs).
 - Image index: figure/diagram embeddings + caption embeddings.
 - Audio/video index: transcript segments + timestamps.
- Ranking layer: modality-specific rerankers + deterministic fusion.
- Answer orchestration: evidence selection + Arabic generation + citation rendering.
- Governance & security: classification, ACL enforcement, logging, audit.

Search foundation

- Use a self-hostable search engine supporting BM25 + vector + hybrid, such as Elastic self-managed or OpenSearch, and build multimodal retrieval pipelines on top (including image/page embeddings).

103

Ingestion and Processing Pipeline

AICA's ingestion must be designed around Saudi document reality: **PDF-first, scan-heavy, bilingual, semi-structured**.

Canonical document object model (DOM)

Every ingested object becomes:

- `asset_id`, `source_system`, `tenant_id`
- `asset_type` : pdf/doc/ppt/image/audio/video
- `language_profile` : Arabic/English/bilingual (probabilistic)
- `security_labels` : classification + ACLs
- `structure` : pages → blocks → (text/table/figure/form fields)
- `evidence_anchors` : stable pointers (page number + bbox coordinates; timestamp ranges)

Layout-aware representation is consistent with Document AI research directions (layout segmentation datasets and models exist precisely because “layout is meaning,” not decoration). ¹⁰⁷

PDF handling (digital vs scanned)

1) Detect PDF type

- If the PDF contains extractable text, treat as “digital PDF” → extract text + layout coordinates.
- If not, or if text extraction confidence is low, treat as “scanned/visual PDF” → rasterize each page image and process as image-first.

2) For scanned/visual PDFs, do not rely on OCR alone

- Run Arabic OCR, but also compute **page-image embeddings** to enable OCR-bypass retrieval (ColPali-like). This is justified by research showing visually rich retrieval benefits from VLM-based embeddings from page images. ⁴

3) Tables, forms, and stamps

- Use layout segmentation and table understanding to extract structured tables where possible (cell grid or key-value pairs), because purely textual OCR output often destroys table semantics.
- Store the rendered table as:
 - a structured representation (rows/cells),
 - a text “table narration” for embeddings,
 - a page region image crop for evidence display.

Embedded images inside PDFs and standalone images

For each figure/diagram/screenshot:

- Extract or crop the image and store as `image_asset_id`.
- Generate:

- image embedding (for image-to-image and text-to-image retrieval),
- Arabic caption/description (generated offline using local VLM/LLM),
- English caption if present/needed for bilingual corpora,
- link back to source PDF page and bbox. Elastic explicitly demonstrates multimodal search (text-to-image/image-to-image) using embeddings and vector search, which validates the feasibility of image retrieval building blocks inside self-managed search stacks. ¹⁰⁸

Audio and video processing

Pipeline: 1) Speech-to-text (ASR) → transcript with timestamps. 2) Segment transcript into retrievable chunks (e.g., 15–45 seconds with overlap). 3) Store segments in text index + keep `time_start`, `time_end`, speaker labels (if diarization). 4) Optionally compute audio embeddings for non-speech audio search, but for enterprise knowledge, transcript retrieval is usually primary.

Whisper is a widely used multilingual ASR model trained on 680k hours; it demonstrates multilingual capabilities including Arabic, but Arabic performance varies by dialect and domain, so AICA must plan for Saudi-specific evaluation and potential fine-tuning. ¹⁰⁹

Retrieval and Ranking Architecture

Query understanding (Arabic-first)

At runtime, AICA should run a lightweight query router that detects:

- language: Arabic vs English vs mixed,
- intent: lookup vs summarize vs compare vs compliance clause,
- evidence need: “text answer” vs “must attach media,”
- modality hint: mentions “صورة/مخطط/جدول/مرفق/خطاب/تعميم/محضر/تسجيل”.

Retrieval strategy (multi-stage, multimodal)

Stage 1: Candidate retrieval

- Text candidates:
 - lexical BM25 over normalized Arabic text + metadata,
 - dense embeddings over Arabic text chunks,
 - hybrid fusion (OpenSearch and other platforms document hybrid vector+lexical retrieval approaches). ¹¹⁰
- Page-image candidates (for scanned/visual PDFs):
 - VLM page embedding retrieval (ColPali-like) over page images.
 - This acts as the robustness layer when OCR is noisy. ¹¹¹
- Image candidates:
 - image embedding retrieval plus caption-text retrieval.
- Audio/video candidates:
 - transcript segment retrieval (dense + lexical).

Stage 2: Reranking

- Arabic reranker (cross-encoder) fine-tuned on Saudi enterprise queries and internal corpora.
- For scanned pages retrieved visually, rerank using:
- query-to-page-image interaction score + OCR text match score.

Stage 3: Evidence consolidation

- Convert the top evidence candidates into:
- cited snippets (text spans),
- page-region highlights (bbox),
- table extracts,
- media attachments.

Response Orchestration

AICA should separate two outputs: 1) **Answer text** (Arabic-first generation): structured, concise, with explicit citations. 2) **Evidence package**: attachments + an evidence map.

Critical safeguards

- Refuse-to-answer behavior when evidence is missing or permissions block access.
- “No citation, no claim” policy: generation prompt must enforce citing anchors.
- “Explainability stub”: provide a short Arabic explanation of evidence basis (“اعتمدتُ على تعميم رقم... الصفحة...”) and show which assets were consulted.

This operationalizes the “traceable decisions” and transparency emphasis in AI ethics principles and aligns with grounding best practices described in cloud grounding docs. ¹¹²

Security, Access Control, and Governance

Permissioning model

Use a strict **document-level and fragment-level ACL**:

- `doc_acl` : users/groups/roles allowed.
- `fragment_acl` : inherit from doc unless overridden.
- Enforce ACL filtering:
- at ingestion (do not index into tenant if no rights),
- at retrieval (filter candidates before reranking),
- at response packaging (do not attach restricted evidence objects).

Competitor documentation emphasizes permission enforcement via connectors and permissions maps; AICA must treat this as mandatory, especially under Saudi regulated expectations. ¹¹³

Cybersecurity alignment

Map AICA controls explicitly to NCA ECC/CCC domains:

- Logging/monitoring: capture retrieval traces, model calls, and access events (consistent with ECC event log/monitoring requirements). ¹¹⁴
- Incident response: define escalation and reporting procedures (ECC includes incident and threat management elements). ¹¹⁴
- Cryptography/key management: encryption at rest/in transit, controlled key lifecycle (ECC includes cryptography requirements). ¹¹⁴
- Cloud/hosting controls: where deployed in cloud-like environments, align with CCC 2-2024 baseline requirements and localization expectations. ³⁶

PDPL considerations

Because PDPL defines “processing” broadly, AICA must design:

- data minimization in logs and traces,
- controlled retention policies,
- anonymization/pseudonymization for evaluation datasets,
- explicit handling of sensitive data and marketing restrictions (PDPL guidance highlights special requirements for sensitive data processing and notes that anonymization removes data from PDPL scope when done correctly, while anonymization itself is still “processing”). ²

Saudi Hosting and Compliance Considerations

Saudi Cloud First policy documentation explicitly states that data in Government Cloud and Commercial Governmental Cloud should be located geographically inside Saudi borders and emphasizes cybersecurity evaluation aligned to NCA requirements. ³⁷ AICA's stated operating model (local hosting, local optimization) is therefore aligned to the spirit of these expectations, but must be executed as:

- tenant-isolated deployments,
- Saudi-controlled keys and admin access,
- auditable infrastructure + application logs.

Arabic Language and Arabic Multimodal Strategy

This is where AICA must be **architecturally opinionated**.

Arabic OCR and document understanding

Observed constraints: Arabic OCR is difficult; systems vary; some cloud OCR supports Arabic, while other popular enterprise OCR services (e.g., Textract) historically exclude Arabic per documentation. ¹¹⁵

AICA strategy

- Use OCR, but treat OCR output as *one signal* among others.
- For visually rich/scanned PDFs, always compute:

- page-image embeddings for retrieval (OCR-bypass),
- OCR text for lexical anchors and snippet display,
- layout segmentation for table/forms.

Donut's argument about OCR error propagation reinforces why OCR-only pipelines are brittle; for AICA, this brittleness is amplified in Arabic. ¹¹⁶

Arabic embeddings and retrieval

Key critique: Multilingual embeddings are useful but do not guarantee Arabic-first results; even multilingual embedding model documentation warns low-resource languages may degrade. ¹¹⁷

AICA approach

- Benchmark at least three embedding families locally: 1) multilingual E5 (strong baseline, cross-lingual potential), ³³
2) an Arabic-specialized encoder family (AraBERT-derived embeddings, or equivalent), justified by Arabic-specific pretraining work, ³²
3) domain-adapted embeddings trained on Saudi enterprise corpora (contrastive learning on internal query-document pairs).
- Use hybrid retrieval: sparse lexical (BM25) + dense semantic.
- Add Arabic normalization and morphology-aware query expansion (light stemming, normalization of orthographic variants, handling diacritics and hamza forms). Arabic's morphological richness is explicitly highlighted in Arabic NLP research; ignoring it causes recall loss. ¹¹⁸

Arabic audio/video

- Use Whisper-class ASR as baseline and evaluate on Saudi dialects; if needed, fine-tune for domain and accent.
- Maintain timestamps for evidence grounding. ¹⁰⁹

Arabic answer generation and citation UX

AICA should adopt a consistent Arabic register suitable for enterprise (formal MSA with controlled use of loanwords). The UX must show citations in a way Arabic readers trust:

- Arabic numerals or Western numerals consistently,
- clear page references,
- right-to-left rendering that does not break snippet ordering.

Because some OCR systems can output incorrect word order for RTL text (vendor support discussions exist), AICA must validate RTL rendering and snippet extraction as a first-class quality dimension. ¹¹⁹

Evaluation and Quality Framework

AICA must implement a multi-layer evaluation stack.

1) OCR quality

- Character error rate (CER), word error rate (WER), plus RTL ordering correctness.
- Stratify by document type: scanned letters, forms, tables, low-quality fax-like scans.

2) Retrieval quality

- Recall@k and nDCG for:
- Arabic text retrieval,
- bilingual retrieval (Arabic query→English doc and vice versa),
- page-image retrieval on scanned PDFs.

3) Grounded answering quality

- Use Ragas-style evaluation to measure faithfulness/groundedness proxies and retrieval quality; adapt prompts for Arabic. ¹²⁰
- Use TruLens-style context relevance and groundedness checks where useful. ⁵⁹

4) Multimodal evidence correctness

- New AICA-specific metric: “Evidence Anchor Accuracy”:
- Does the cited bbox/time segment contain the answer claim?
- Can a reviewer verify in <15 seconds?

5) Production observability

- Use self-hostable tracing and evaluation tooling such as Phoenix (OpenTelemetry-based) to capture retrieval traces and evaluate drift. ⁶⁰

Operations, Monitoring, and Improvement Loop

AICA should run two loops:

- **Offline loop:** curated test sets + regression evaluation across OCR, retrieval, and grounded QA.
 - **Online loop:** user feedback + flagged “low evidence sufficiency” answers feed back into:
 - ingestion fixes (better OCR configs, layout models),
 - retrieval tuning (fusion weights, reranker training),
 - policy updates (refuse thresholds, safety rules).
-

Decisioning and Delivery Plan

Technical Assumptions

These are explicit assumptions used to make concrete recommendations:

- AICA will be deployed as a **tenant-isolated Saudi-local system** (private cloud or on-prem), consistent with Cloud First localization expectations for government cloud contexts and with sovereign positioning. ³⁷
- The majority of high-value knowledge for first deployments will be **PDF-heavy**, with a non-trivial share of scanned PDFs and embedded images. (Assumption based on described scenario, not an observed market statistic.)
- AICA can run local GPU inference for embeddings and VLM tasks (needed for ColPali-like page embeddings and for local OCR/VLM pipelines).
- Customers require permission-aware retrieval; connectors must ingest ACLs or equivalent security mappings (aligned with industry best practice). ¹¹³

Build vs Buy Analysis

This section recommends what to build vs integrate, under Saudi sovereignty constraints.

OCR

- Buy (on-prem capable) for baseline enterprise accuracy + support: ABBYY explicitly supports Arabic and offers end-to-end Arabic recognition for low-quality images. ³¹
- Build/augment: an internal OCR evaluation and post-correction pipeline (Arabic normalization, dictionary/domain lexicons, RTL ordering validation). Arabic OCR review literature supports that preprocessing/segmentation materially affect quality, and Arabic properties increase complexity. ²⁷

Position: Buy the engine where feasible, but build the Saudi Arabic quality layer and evaluation.

Document parsing and layout

- Build using open research and open-source components:
- layout segmentation models inspired by DocLayNet (dataset exists for training/evaluation), ¹²¹
- optionally LayoutLMv3-like approaches for extracting structure and aligning text/image/layout. ¹²²

Position: Build core DOM + connectors; use open models as modules.

Vision-first page retrieval (critical differentiator)

- Build: AICA should own this subsystem because it is the core differentiator for scanned Arabic PDFs.
- Use research-based models: ColPali provides an existence proof that page-image embeddings and late interaction outperform standard pipelines. ¹²³

Position: Build in-house capability with open checkpoints where licensing allows; treat as “AICA secret sauce.”

ASR

- Build locally using Whisper-class models and fine-tuning capability; Whisper is open and well documented. ¹²⁴

Position: Build and fine-tune; avoid cloud ASR dependencies.

Embeddings

- Start with strong open models (multilingual E5 baseline), then build domain-adapted Arabic embeddings.
- Arabic-specialized models like AraBERT justify Arabic-specific modeling; evaluate both. ¹²⁵

Position: Build evaluation + domain adaptation; integrate open models.

Vector storage / search

- Buy vs build is misguided here: use mature self-host foundations.
- Elastic self-managed and OpenSearch vector+hybrid capabilities are documented; choose one and standardize. ¹²⁶

Position: Integrate, do not build.

Reranking

- Build: Arabic reranker models fine-tuned on enterprise relevance judgments. **Position:** Build.

Orchestration

- Use open frameworks for acceleration (LlamaIndex multi-modal guidance exists; LangChain retrieval docs exist), but build an internal orchestration layer with stable interfaces to avoid lock-in. ¹²⁷

Position: Hybrid—use frameworks as libraries, own the platform.

Evaluation tooling

- Integrate open tooling: Ragas, TruLens, Phoenix are documented as evaluation/tracing tools. ¹²⁸

Position: Integrate and extend with Arabic multimodal evals.

Monitoring / observability

- Use Phoenix/OpenTelemetry-like approaches for tracing; build Saudi-compliant log retention and redaction. ¹²⁹

Position: Integrate + build compliance layer.

Safety / governance tooling

- Build: policy engine, refusal logic, audit dashboards, classification enforcement, redaction for logs.
- Align with AI ethics principles for lifecycle governance and transparency. ⁵⁶

Recommended Product Features and Differentiators

AICA should differentiate on features that are hard for cloud copilots to replicate under sovereign constraints:

1) Arabic-first evidence UX

- Page-region highlighting for PDFs.
- Table extraction with “cell-level citation.”
- Consistent Arabic citation formatting and RTL-safe snippet rendering.

2) Scanned-document robustness

- OCR + page-image retrieval (ColPali-like) + layout-aware parsing.
- “Explain why this scan was used” transparency.

3) Bilingual bridging (Arabic↔English)

- Arabic question retrieves English annex and vice versa, with explicit translation of citations.

4) Sovereign-by-design deployment

- All models (generation, embeddings, ASR, VLM retriever) run locally in Saudi infrastructure.
- Customer-controlled keys and admin boundaries (aligned to localization expectations). ¹³⁰

5) Audit-grade governance

- Retrieval trace export, evidence map, and PDPL-friendly logging.

MVP Recommendation

Build first (MVP)

- PDF ingestion (digital + scanned)
- Arabic OCR + quality evaluation harness
- Page-image embedding retrieval for PDFs (scanned/visual)
- Hybrid Arabic text retrieval (BM25 + dense)
- Arabic answer generation with citations + page-level anchors
- Permission-aware retrieval (doc-level ACLs at minimum)
- Minimal admin console for corpus ingestion and evaluation snapshots

Postpone

- Video understanding beyond transcript-based retrieval
- Diagram reasoning beyond captioning + retrieval
- Agentic workflow automation (focus first on correctness and trust)

This sequencing is justified because PDF + scans are the dominant “Saudi document reality” in the scenario and because VLM-based retrieval yields disproportionate value for visually rich documents. ⁴

Roadmap Recommendation

Phase 1

- MVP features above.
- Build Arabic evaluation datasets and establish regression gates using Ragas-style and Phoenix-style instrumentation. ¹³¹

Phase 2

- Advanced document AI:
 - table structure extraction at higher accuracy,
 - form field extraction for common Saudi government templates,
 - improved Arabic reranker and domain-adapted embeddings.
- Audio ingestion and timestamp citations (Whisper baseline + tuning). ¹⁰⁹

Phase 3

- Full multimodal evidence orchestration:
 - image reasoning for charts/diagrams,
 - video “key moment” indexing,
 - human-in-the-loop workflows for high-stakes answers,
 - automated governance reports for auditors.

Trade-offs and Alternatives

Option A: OCR-first multimodal (traditional)

- Pros: simpler; uses text retrieval infrastructure.
- Cons: fails on Arabic scans; loses layout semantics; higher hallucination risk because evidence is degraded.
- Not recommended as primary approach for AICA. ⁶¹

Option B: Vision-first page retrieval + OCR (recommended)

- Pros: robust to OCR failures; captures layout; aligned with ColPali/VisRAG direction for visually rich docs; produces better evidence objects.
- Cons: higher GPU/storage cost; requires careful engineering for page rendering and multi-vector indexing.
- Recommended because Saudi enterprise value sits in PDFs/scans. ¹¹¹

Option C: OCR-free document understanding (Donut-like)

- Pros: avoids OCR error propagation; potentially language-flexible.
- Cons: training/fine-tuning for enterprise Arabic documents is non-trivial; riskier for MVP.
- Consider for specific high-value document types (invoices/forms) later. ¹³²

Final Strategic Recommendation

AICA should bet on a defensible leadership position: **Arabic-first Multimodal RAG with sovereign deployment and audit-grade evidence UX.**

This is not a generic RAG product. It is a Saudi enterprise-grade assistant designed to meet:

- the operational reality of scanned and media-heavy Arabic knowledge,
- the regulatory expectations implied by PDPL’s broad processing scope, ²
- cybersecurity governance expectations reflected in ECC/CCC control frameworks, ⁵⁷
- and responsible AI expectations emphasizing transparency, traceability, privacy/security-by-design across the lifecycle. ⁵⁶

Competitor analysis shows a clear opening:

- Some major platforms explicitly do not index non-English content (Amazon Q Business), which is incompatible with Arabic-first enterprise knowledge assistants. ⁷⁰
- Others support Arabic at the prompt level but are not positioned as sovereign, Arabic-first, multimodal evidence engines.
- Enterprise search vendors with sovereign deployment options exist, but Arabic-first multimodal performance is not assured without AICA-level specialization. ¹³³

Therefore, AICA’s “win condition” is **trust**: measurable groundedness, citations, and evidence objects in Arabic, delivered within Saudi-controlled infrastructure.

Prioritized Decision Table

Capability	Priority	Build vs Buy	Expected competitive impact	Implementation complexity	Arabic-specific risk level
Canonical multimodal document object model (pages, bboxes, tables, links)	Must-have	Build	Very high (enables audit-grade UX)	High	Medium
Arabic OCR engine (baseline)	Must-have	Buy (on-prem) + integrate	High	Medium	High (quality variance)
Arabic OCR quality harness (CER/WER/RTL order) + post-correction	Must-have	Build	High	Medium	High
Page-image embedding retrieval for PDFs (ColPali-like)	Must-have	Build (core IP)	Very high (scan robustness)	High	Medium

Capability	Priority	Build vs Buy	Expected competitive impact	Implementation complexity	Arabic-specific risk level
Hybrid Arabic retrieval (BM25 + dense vectors)	Must-have	Build on existing search	High	Medium	Medium
Arabic embeddings baseline (multilingual E5 + Arabic-specialized evaluation)	Must-have	Integrate + evaluate	High	Medium	Medium ¹³⁴
Arabic reranking model tuned on enterprise relevance	Must-have	Build	High	High	Medium
Permission-aware retrieval (ACL ingestion + query-time filtering)	Must-have	Build	Very high (enterprise trust)	High	Medium
Evidence packaging: page-region highlights + snippet citations	Must-have	Build	Very high (differentiator)	High	Medium
Self-host search foundation (Elastic/OpenSearch)	Must-have	Buy/Integrate (self-host)	Medium	Medium	Low ¹³⁵
LLM orchestration layer (gating, prompting, refusal, evidence sufficiency)	Must-have	Build	High	Medium	Medium
Arabic ASR for audio/video + timestamp evidence	Should-have	Build (open models)	Medium/High	High	High ¹³⁶
Image/diagram captioning (Arabic) for figures/screenshots	Should-have	Build + integrate	Medium	Medium	Medium
Evaluation platform (Ragas + Phoenix + custom Arabic multimodal metrics)	Should-have	Integrate + extend	High (quality defensibility)	Medium	Medium ¹³⁷

Capability	Priority	Build vs Buy	Expected competitive impact	Implementation complexity	Arabic-specific risk level
Governance dashboard (audit export, policy mapping, incident review)	Should-have	Build	High (government readiness)	Medium	Low/ Medium
OCR-free doc understanding for select templates (Donut-like)	Later-phase	Build (R&D)	Medium	High	High ¹³²
Video “key moment” indexing (beyond transcripts)	Later-phase	Build	Medium	High	Medium
Agentic workflow automation (actions, ticketing, approvals)	Later-phase	Build	Medium	High	Low/ Medium
Continuous domain adaptation (embeddings + reranker + ASR tuning)	Later-phase	Build	High long-term	High	Medium/ High

¹ ²¹ ¹⁰⁴ <https://openreview.net/forum?id=zG459X3Xge>

<https://openreview.net/forum?id=zG459X3Xge>

² ¹² ¹⁴ ¹⁹ <https://dgp.sdaia.gov.sa/wps/wcm/connect/f579bc32-fda8-47bd-bc6f-66b8cb77985c/ENG-Guide%2Bto%2Bthe%2Bsaudi%2BPDP%2Blaw%2Bfor%2Bcontrollersprocessors.pdf?MOD=AJPERES>

<https://dgp.sdaia.gov.sa/wps/wcm/connect/f579bc32-fda8-47bd-bc6f-66b8cb77985c/ENG-Guide%2Bto%2Bthe%2Bsaudi%2BPDP%2Blaw%2Bfor%2Bcontrollersprocessors.pdf?MOD=AJPERES>

³ ³⁵ ⁵⁷ ¹¹⁴ https://cdn.nca.gov.sa/api/files/public/upload/86e09090-44e4-481f-bc28-355673607654_ECC--2024-EN.pdf

https://cdn.nca.gov.sa/api/files/public/upload/86e09090-44e4-481f-bc28-355673607654_ECC--2024-EN.pdf

⁴ ⁵ ¹¹ ¹³ ¹⁶ ²⁰ ²² ⁴⁷ ⁴⁹ ⁶⁵ ⁸⁰ ⁸² ¹⁰⁵ ¹¹¹ ¹²³ <https://arxiv.org/abs/2407.01449>

<https://arxiv.org/abs/2407.01449>

⁶ ⁸³ ⁹⁷ ¹⁰³ ¹²⁶ ¹³⁵ <https://www.elastic.co/docs/deploy-manage/deploy/self-managed>

<https://www.elastic.co/docs/deploy-manage/deploy/self-managed>

⁷ ²⁴ ¹²² <https://arxiv.org/abs/2204.08387>

<https://arxiv.org/abs/2204.08387>

8 110 <https://docs.opensearch.org/3.0/tutorials/vector-search/neural-search-tutorial/>
<https://docs.opensearch.org/3.0/tutorials/vector-search/neural-search-tutorial/>

9 27 46 48 61 115 <https://www.sciencedirect.com/org/science/article/pii/S1526149222002557>
<https://www.sciencedirect.com/org/science/article/pii/S1526149222002557>

10 40 76 <https://www.servicenow.com/docs/r/platform-administration/ai-search/ais-rag.html>
<https://www.servicenow.com/docs/r/platform-administration/ai-search/ais-rag.html>

15 17 44 54 67 74 89 <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>
<https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-architecture>

18 39 88 <https://support.microsoft.com/en-us/office/supported-languages-for-microsoft-365-copilot-94518d61-644b-4118-9492-617eea4801d8>
<https://support.microsoft.com/en-us/office/supported-languages-for-microsoft-365-copilot-94518d61-644b-4118-9492-617eea4801d8>

23 116 132 <https://arxiv.org/abs/2111.15664>
<https://arxiv.org/abs/2111.15664>

25 <https://github.com/DS4SD/DocLayNet>
<https://github.com/DS4SD/DocLayNet>

26 <https://arxiv.org/abs/2007.00398>
<https://arxiv.org/abs/2007.00398>

28 <https://link.springer.com/article/10.1007/s42001-021-00149-1>
<https://link.springer.com/article/10.1007/s42001-021-00149-1>

29 <https://docs.cloud.google.com/vision/docs/languages>
<https://docs.cloud.google.com/vision/docs/languages>

30 <https://learn.microsoft.com/en-us/azure/ai-services/computer-vision/overview-ocr>
<https://learn.microsoft.com/en-us/azure/ai-services/computer-vision/overview-ocr>

31 119 <https://support.abbyy.com/hc/en-us/articles/360016365800-OCR-for-Arabic-and-Farsi>
<https://support.abbyy.com/hc/en-us/articles/360016365800-OCR-for-Arabic-and-Farsi>

32 62 125 <https://aclanthology.org/2020.osact-1.2/>
<https://aclanthology.org/2020.osact-1.2/>

33 <https://arxiv.org/abs/2402.05672>
<https://arxiv.org/abs/2402.05672>

34 38 109 124 <https://cdn.openai.com/papers/whisper.pdf>
<https://cdn.openai.com/papers/whisper.pdf>

36 https://cdn.nca.gov.sa/api/files/public/upload/6d5408a3-d8e6-4e96-963b-2c7198e5b7c2_CCC-2-2024-EN-.pdf
https://cdn.nca.gov.sa/api/files/public/upload/6d5408a3-d8e6-4e96-963b-2c7198e5b7c2_CCC-2-2024-EN-.pdf

37 50 130 https://www.mcit.gov.sa/sites/default/files/cloud_policy_en.pdf
https://www.mcit.gov.sa/sites/default/files/cloud_policy_en.pdf

41 70 101 <https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/supported-languages.html>
<https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/supported-languages.html>

42 86 <https://ibm.github.io/watsonx-workshop-genai/lab-generative-ai-and-rag/>
<https://ibm.github.io/watsonx-workshop-genai/lab-generative-ai-and-rag/>

43 73 <https://docs.glean.com/administration/management/features/glean-language-support>
<https://docs.glean.com/administration/management/features/glean-language-support>

45 68 <https://docs.cloud.google.com/generative-ai-app-builder/docs/languages-locales>
<https://docs.cloud.google.com/generative-ai-app-builder/docs/languages-locales>

51 90 100 <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/grounding/overview>
<https://docs.cloud.google.com/vertex-ai/generative-ai/docs/grounding/overview>

52 75 <https://www.servicenow.com/docs/r/zurich/platform-administration/ai-search/now-assist-ais.html>
<https://www.servicenow.com/docs/r/zurich/platform-administration/ai-search/now-assist-ais.html>

53 56 106 112 https://dgp.sdaia.gov.sa/wps/wcm/connect/4c56ed1c-1b82-447d-ac29-638f5f99c12e/ai-principles-EN.pdf?CACHEID=ROOTWORKSPACE-4c56ed1c-1b82-447d-ac29-638f5f99c12e-p3k51U9&CONVERT_TO=url&MOD=AJPERES
https://dgp.sdaia.gov.sa/wps/wcm/connect/4c56ed1c-1b82-447d-ac29-638f5f99c12e/ai-principles-EN.pdf?CACHEID=ROOTWORKSPACE-4c56ed1c-1b82-447d-ac29-638f5f99c12e-p3k51U9&CONVERT_TO=url&MOD=AJPERES

55 64 71 113 <https://docs.glean.com/connectors/about>
<https://docs.glean.com/connectors/about>

58 131 <https://docs.ragas.io/en/stable/>
<https://docs.ragas.io/en/stable/>

59 https://www.trulens.org/getting_started/core_concepts/rag_triad/
https://www.trulens.org/getting_started/core_concepts/rag_triad/

60 129 <https://github.com/arize-ai/phoenix>
<https://github.com/arize-ai/phoenix>

63 66 <https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-overview>
<https://learn.microsoft.com/en-us/copilot/microsoft-365/microsoft-365-copilot-overview>

69 92 <https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/connectors-list.html>
<https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/connectors-list.html>

72 <https://docs.glean.com/get-started/build/about-self-hosted>
<https://docs.glean.com/get-started/build/about-self-hosted>

77 78 79 102 <https://docs.coveo.com/en/n9de0370/>
<https://docs.coveo.com/en/n9de0370/>

81 95 133 <https://www.chapsvision.com/en-us/platform/sinequa/>
<https://www.chapsvision.com/en-us/platform/sinequa/>

84 <https://www.elastic.co/search-labs/blog/multimodal-image-retrieval-with-roboflow>
<https://www.elastic.co/search-labs/blog/multimodal-image-retrieval-with-roboflow>

85 120 128 137 <https://arxiv.org/abs/2309.15217>
<https://arxiv.org/abs/2309.15217>

87 <https://mea.newsroom.ibm.com/sdaia-launches-allam-on-watsonx>
<https://mea.newsroom.ibm.com/sdaia-launches-allam-on-watsonx>

⁹¹ <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/grounding/grounding-with-vertex-ai-search>
<https://docs.cloud.google.com/vertex-ai/generative-ai/docs/grounding/grounding-with-vertex-ai-search>

⁹³ <https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/connector-concepts.html>
<https://docs.aws.amazon.com/amazonq/latest/qbusiness-ug/connector-concepts.html>

⁹⁴ <https://docs.coveo.com/en/2976/>
<https://docs.coveo.com/en/2976/>

⁹⁶ ¹⁰⁸ <https://www.elastic.co/search-labs/blog/multimodal-search-siglip-2-elasticsearch>
<https://www.elastic.co/search-labs/blog/multimodal-search-siglip-2-elasticsearch>

⁹⁸ <https://docs.opensearch.org/latest/search-plugins/>
<https://docs.opensearch.org/latest/search-plugins/>

⁹⁹ <https://docs.opensearch.org/latest/vector-search/vector-search-techniques/index/>
<https://docs.opensearch.org/latest/vector-search/vector-search-techniques/index/>

¹⁰⁷ ¹²¹ <https://arxiv.org/abs/2206.01062>
<https://arxiv.org/abs/2206.01062>

¹¹⁷ ¹³⁴ <https://huggingface.co/intfloat/multilingual-e5-large>
<https://huggingface.co/intfloat/multilingual-e5-large>

¹¹⁸ <https://arxiv.org/abs/2003.00104>
<https://arxiv.org/abs/2003.00104>

¹²⁷ https://developers.llamaindex.ai/python/framework/module_guides/models/multi_modal/
https://developers.llamaindex.ai/python/framework/module_guides/models/multi_modal/

¹³⁶ https://www.isca-archive.org/interspeech_2025/wang25_interspeech.pdf
https://www.isca-archive.org/interspeech_2025/wang25_interspeech.pdf